

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is that the server was overwhelmed by a large number of TCP SYN requests. The logs show many SYN packets coming from an unfamiliar spoofed IP address. This event could be a SYN flood attack.

Section 2: Explain how the attack is causing the website to malfunction

A user's device sends a SYN request to the server to begin a connection. The server responds with a SYN-ACK and waits for the final ACK to complete the TCP three-way handshake. During the attack, the malicious actor sends many SYN requests but never sends the final ACK. This leaves many half-open connections open on the server until its resources become overwhelmed. As a result, legitimate users are unable to establish new connections and cannot access the website.